Type MS · Judge whether data support a claim and spot their limits

Answer 1 of 8 · Question 7 part c.iii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.iii in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Judge whether data support a claim and spot their limits

Answer 1 of 8 · Question 7 part c.iii · 4 marks

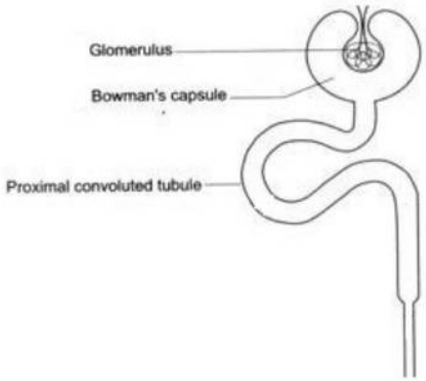
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.iii in this mark scheme extract

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Type MS · Judge whether data support a claim and spot their limits

Answer 2 of 8 · Question 9 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

9 (a) (i)	• evidence of 23500 – 4000 (= 19500) (1) • 19500 ÷ 4000 (x 100)(1) • 488(%)	Allow full marks for correct final answer. Allow 487.5(%) Allow ecf if incorrect answer from mp1 – max 2 marks	3
(ii)	other lifestyle factors/genetic predisposition could have an influence/named lifestyle factor(1)	Allow gender/age	1
(b)	A diagram similar to:  	One mark for diagram. One mark for each correct label.	4
(c)	any four from • ultrafiltration does not take place(1) • in glomerulus(1) • less/no waste products/named waste product/water/glucose pass through to tubule(1) • greater volume of waste in blood/more water in blood(1) • glucose/ions not reabsorbed(1) • glucose found in urine(1) OR any four from • blood not filtered effectively/ultrafiltration ineffective(1) • in glomerulus (1)		4

Type MS · Judge whether data support a claim and spot their limits

Answer 2 of 8 · Question 9 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

	• larger molecules/named larger molecules pass through to tubule/Bowmans capsule(1) • not all glucose/ions reabsorbed(1) • glucose/protein found in urine/urine more concentrated(1)		

Total for Question 9 = 12 marks

Type MS · Judge whether data support a claim and spot their limits

Answer 3 of 8 · Question 7 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number				
7 (a)	(i)	• axes labelled with units; • correct way round and suitable scale; • correct plots; • suitable line drawn;		4
	(ii)	• only not working correctly when rate falls below 100; • data/year not shown when that occurs/already below 100;		2
	(iii)	• 2021/2022; • rate falling at 5-10 units per annum/similar values each year;		2
	(iv)	• not very accurate/results vary/fluctuate; • earlier data showed improvement in function; • could happen again/results may change;		3
(b)		any three from • protein broken down into urea/deamination; • toxic; • needs to be excreted; • incorrectly functioning kidney can't remove it all if there is a lot; • less protein less urea;		3

Total 14

Type MS · Judge whether data support a claim and spot their limits

Answer 3 of 8 · Question 7 part a.iv · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

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Type MS · Judge whether data support a claim and spot their limits

Answer 4 of 8 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
6	(a)	(i)	Increase/goes up;	allow equivalent	1
		(ii)	Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;;	Allow one mark for reference to lifestyles only	2
	(b)		Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed; Increased risk of (lung/respiratory) infection/coughing;		1 1
	(c)	(i)	<table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table>	Allow for SA:V ratio 6:3 ECF	3
		(ii)	Smaller surface area to volume ratio (in lungs with emphysema); Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide; Less oxygenated blood/less oxygen transferred (to cells); Less (aerobic) respiration;		Max 3

Type MS · Judge whether data support a claim and spot their limits

Answer 4 of 8 · Question 6 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Judge whether data support a claim and spot their limits

Answer 5 of 8 · Question 7 part a.iii · 2 marks

2020 JAN · Paper 01 · find part a.iii in this mark scheme extract

Question number					
7	(a)	(i)	• 67.5 ÷ 48.3 (1) • 1:1.4 (1)	Full marks for correct final answer	2
		(ii)	Any two from • more males than females are overweight in all countries (1) • lowest percentage of people between the ages of 18-24 in all countries are overweight (1) • increased in all countries 18 - 64 (1)		2
		(iii)	sample size too small (1) not all overweight people identified/sample not representative (1)		2
	(b)	(i)	Any three from: • pancreas (1) • releases insulin (1) • reference to the liver taking up glucose (1) • glucose converted to glycogen (1)	R sugar R breaks down glucose	3
		(ii)	Any two from: • high blood glucose increases the concentration of the blood (1) • water moves out of cells (1) • by osmosis (1) • excess water in blood excreted in urine (1)		2

Total for Question 7 = 11 marks

Type MS · Judge whether data support a claim and spot their limits

Answer 6 of 8 · Question 3 part b.iii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

3	(a)	(i)	A (cornea and lens); B is incorrect as the iris does not play a role in focusing light C is incorrect as the iris does not play a role in focusing light D is incorrect as the pupil does not play a role in focusing light		1
	(b)	(i)	Points as follows: 150:2.7 200:3.2 250:2.3;		1
Clip (b)(i) with (b)(ii) and (b)(iii)		(ii)	Line of best fit missing out point at 150:2.7	Ignore lines that extended beyond points	1
		(iii)	150 lux/150:2.7/2.7 diameter;		1
			does not follow the pattern of other points/trend/not on the line of best fit;		1
		(iv)	As light intensity increases diameter decreases;	ORA	1
	(c)		Idea that distant/far away objects are blurred/difficult to see/ORA; as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much;	Reject can't see	1 1

Total question 3 = 8 marks

Type MS · Judge whether data support a claim and spot their limits

Answer 6 of 8 · Question 3 part b.iii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Judge whether data support a claim and spot their limits

Answer 7 of 8 · Question 7 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
7 (a)	(i)	• axes labelled with units; • correct way round and suitable scale; • correct plots; • suitable line drawn;		4
	(ii)	• only not working correctly when rate falls below 100; • data/year not shown when that occurs/already below 100;		2
	(iii)	• 2021/2022; • rate falling at 5-10 units per annum/similar values each year;		2
	(iv)	• not very accurate/results vary/fluctuate; • earlier data showed improvement in function; • could happen again/results may change;		3
(b)		any three from • protein broken down into urea/deamination; • toxic; • needs to be excreted; • incorrectly functioning kidney can't remove it all if there is a lot; • less protein less urea;		3

Total 14

Type MS · Judge whether data support a claim and spot their limits

Answer 7 of 8 · Question 7 part a.ii · 2 marks

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Type MS · Judge whether data support a claim and spot their limits

Answer 8 of 8 · Question 6 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

6 (a) (i)	2500 males; $\frac{2500 \times 0.8}{100}$; = 20;		ecf if use 5000 as starting figure 40 scores 2 marks any number×0.8=1	3
(ii)	small number/ not enough tested;			1
(iii)	• many more males than females with condition; • show that it is sex-linked; • to X chromosome; • females are XX /males one X chromosome/XY • females can be carriers(so fewer)/males cannot be carriers(so more); • (relatively) small number (in whole population)/small chance of being colour blind; • indicates that it is recessive;			5
(b)	parents	$X^BY \times X^BX^b$;		4
	gametes	X^B Y X^B X^b;		
	offspring genotypes	$X^B X^B$ X^BX^b X^BY X^bY;		
	offspring phenotypes	normal carrier/ disease normal red-green;		
		female female male male	If parents are incorrect ecf for mp 2,3 and 4 If X and Y are not used- max of 1 mark	
				Total = 13

Type MS · Judge whether data support a claim and spot their limits

Answer 8 of 8 · Question 6 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks